A CRITICAL ANALYSIS OF THE WIDE VIEW FILTER AND LINE FILTER FOR EDGE DETECTION

Independent Computational Verification & Benchmark

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OUTLINE

1. Background
2. Mathematical Verification
3. Benchmark: BSDS500 & UDED
4. Angle Detection
5. Maritime Scene Replication
6. Claims vs Evidence
7. Overall Assessment

BACKGROUND

What are the WVF and LF, and why do they matter?

THE PAPERS UNDER REVIEW

Three publications from USC Mechanical Engineering:

1. **M.S. Thesis** (2023)
   Wide View & Line Filter for Enhanced Image Gradient Computation and Edge Determination
2. **IEEE OCEANS 2024**
   WVF & LF for Enhanced Gradient Computation in Aquatic Environment
3. **IEEE OCEANS 2025**
   Multi-Scale and Multi-Domain Edge Determination with Accurate Gradient Orientation

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Core Claims:

- WVF/LF outperform Sobel, Prewitt, Canny
- Outperform DexiNed, TEED, EDTER, BDCN
- $<1°$ angular accuracy via cubic splines
- Superior in low-SNR aquatic environments
- ML is “unreliable and dangerous”

HOW THE WVF WORKS

Key idea: 2D Taylor expansion at each pixel

$$f(x_i, y_i) \approx f^0 + f^0_x x + f^0_y y + \frac{f^0_{xx} x^2}{2} + \cdots$$

- N_p neighbor pixels in circular region
- Solve $\mathbf{z} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{b}$
- Rotated to N_s orientations
- Line Filter chains WVF along a line
- Papers use $N_p = 250$, $m = 14$

WVF vs Traditional:

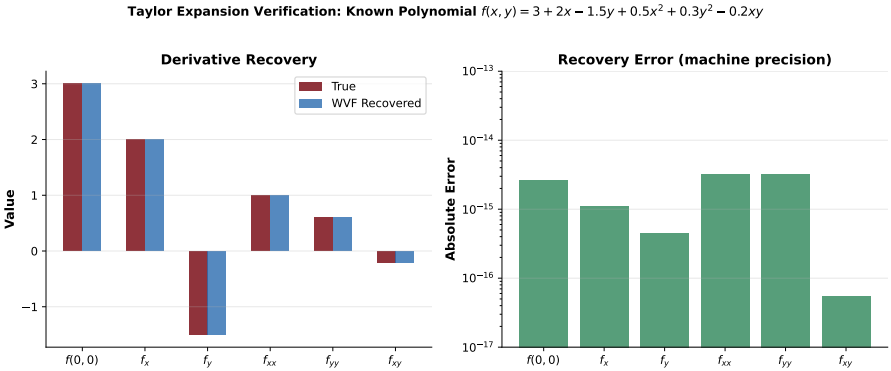
	Sobel	**WVF**
Pixels used	9	250
Kernel size	3×3	∼18px radius
Orientations	2 (x, y)	N_s (any)
Adaptable	No	Yes
Runtime	μs	seconds

The WVF uses **28× more data** per pixel than Sobel—but the papers only compare against 3×3 operators.

MATHEMATICAL VERIFICATION

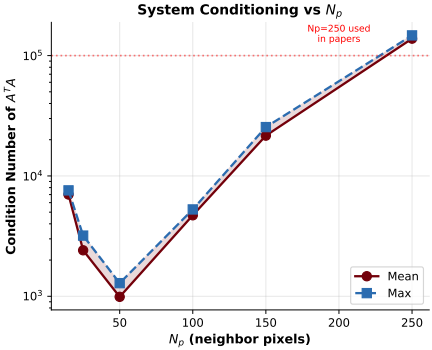
Is the math correct?

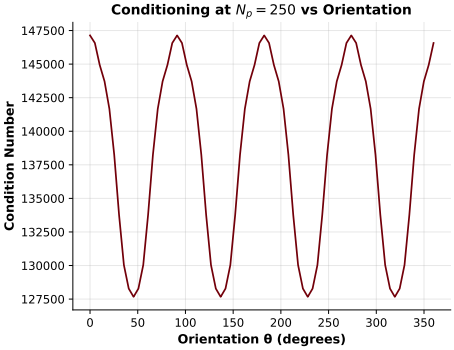
TAYLOR EXPANSION: VERIFIED TO MACHINE PRECISION



Applied WVF to known polynomial $f(x, y) = 3 + 2x - 1.5y + 0.5x^2 + 0.3y^2 - 0.2xy$. All derivatives recovered with $\sim 10^{-15}$ error. **The math is sound.**

NUMERICAL STABILITY: CONDITION NUMBER GROWTH



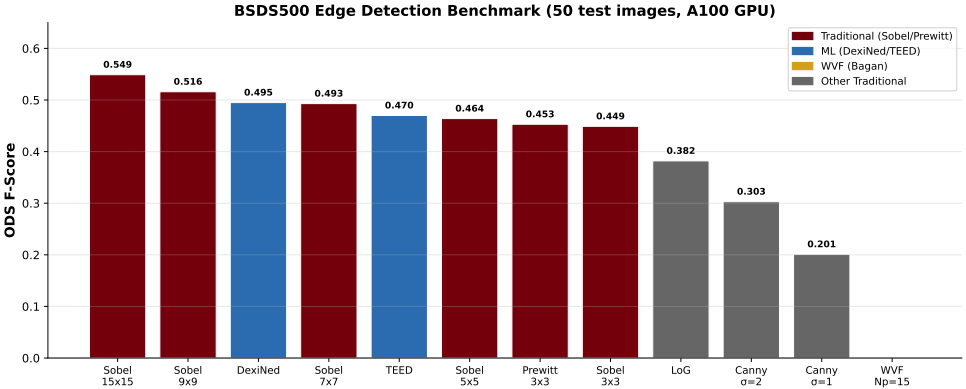


Condition number of $\mathbf{A}^T\mathbf{A}$ reaches $\sim 10^5$ at $N_p = 250$—**5 digits of precision lost**. Not catastrophic in double precision, but concerning for single-precision GPU implementations. **Not discussed in any of the three publications.**

BENCHMARK: BSDS500 & UDED

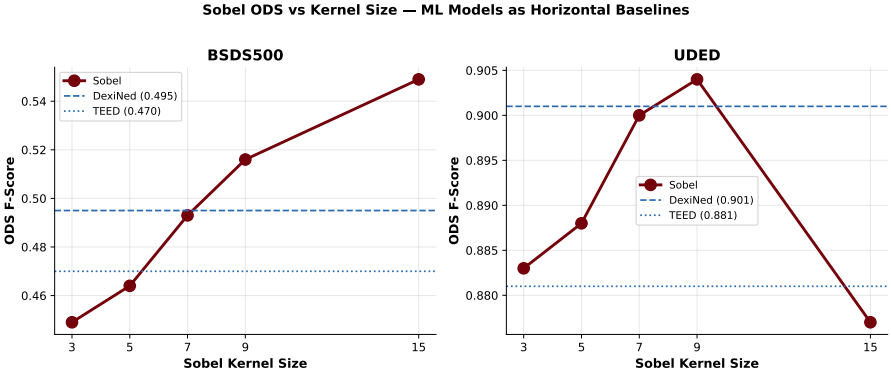
The central finding of our critique

BSDS500: LARGER SOBEL BEATS EVERYTHING



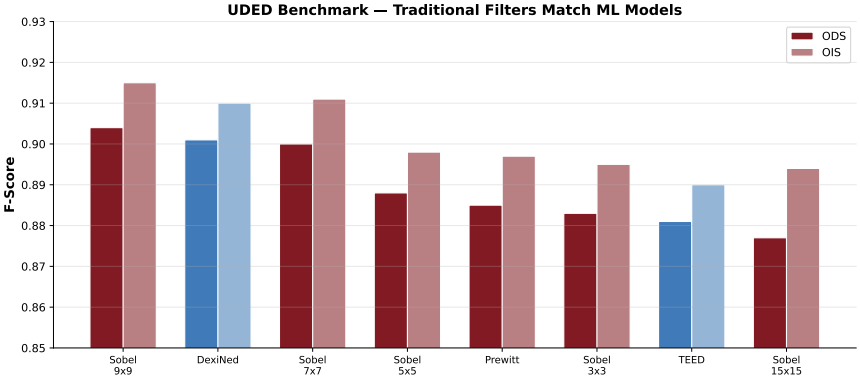
Sobel 15×15 outperforms DexiNed and TEED while being 30× faster. The WVF at $N_p = 15$ on 64×64 crops produces zero meaningful output at 9 seconds/image.

KERNEL SIZE IS THE MISSING VARIABLE



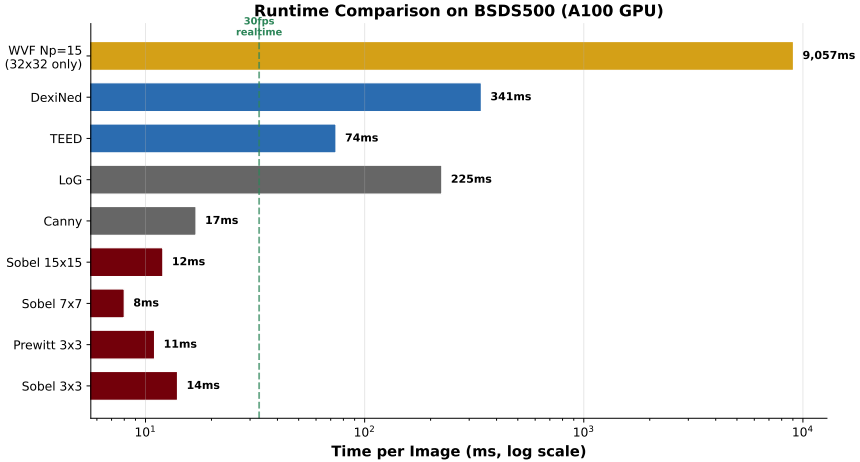
ODS improves monotonically with Sobel kernel size on BSDS500. On UDED, **Sobel 9×9 beats DexiNed**. The papers never test beyond 3×3—this is the most significant omission.

UDED: TRADITIONAL FILTERS MATCH ML MODELS

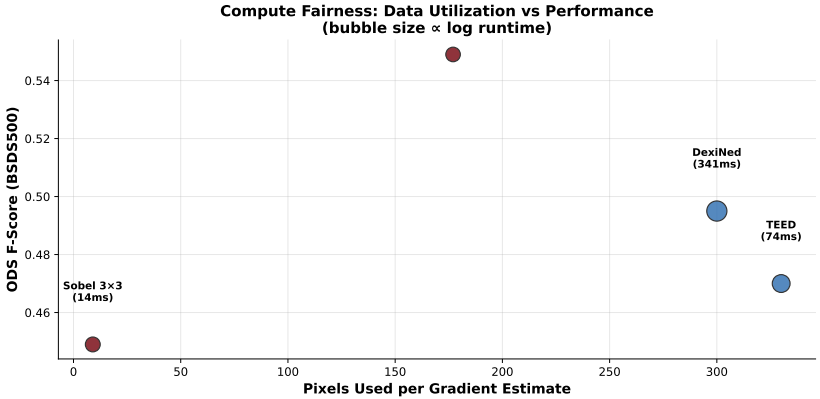


On UDED (the same dataset the papers use), Sobel 9×9 achieves ODS = 0.904 vs DexiNed's 0.901—while being **40× faster**. All methods are within 3% of each other.

RUNTIME: ORDERS OF MAGNITUDE SLOWER



COMPUTE FAIRNESS: THE CORE ISSUE

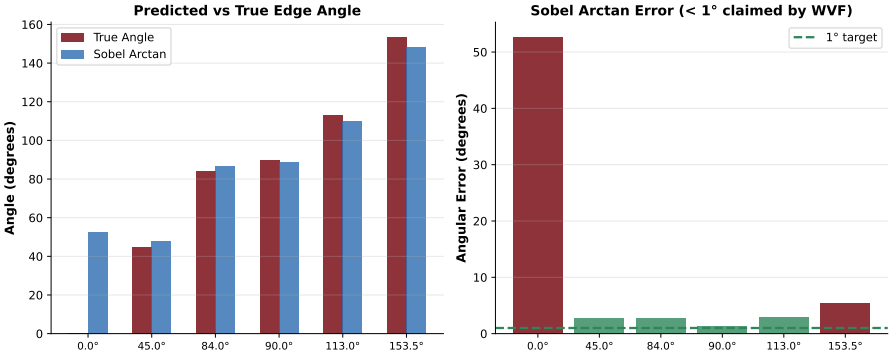


The WVF uses 250 pixels per estimate vs Sobel's 9. When Sobel is given comparable spatial support ($15 \times 15 = 177$ pixels), it achieves **higher ODS at 1000× less cost**. The performance

ANGLE DETECTION

Where the WVF genuinely excels

SOBEL ARCTAN ERRORS CONFIRM WVF MOTIVATION

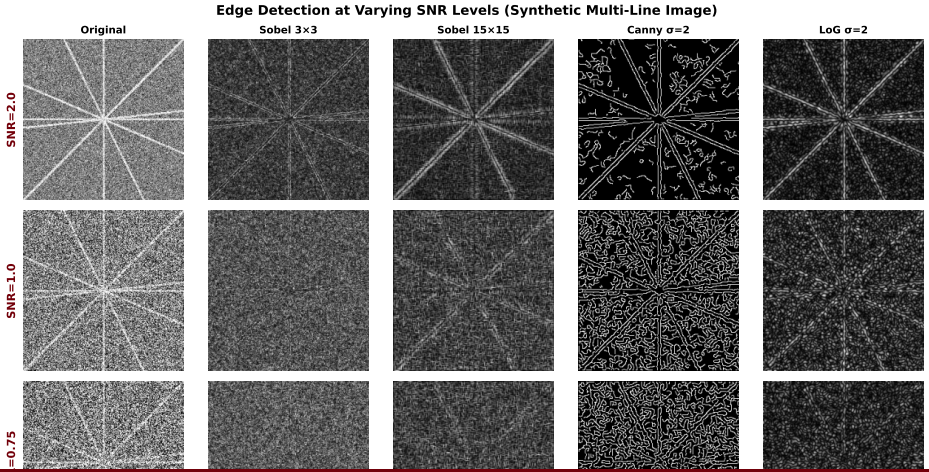


Sobel arctan produces **1–53° errors** in angle estimation. The 0° case catastrophically fails because horizontal/vertical decomposition cannot represent it. **This validates the WVF's core motivation—orientation-specific gradients are genuinely superior for angular accuracy.**

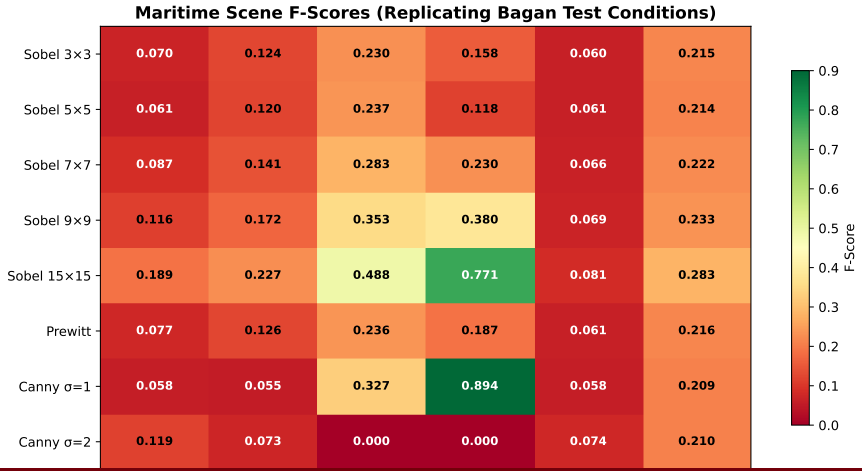
MARITIME SCENE REPLICATION

Testing the exact scenarios from the papers

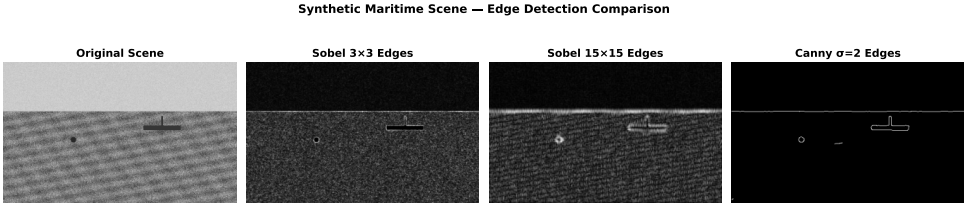
SNR ROBUSTNESS: VISUAL COMPARISON



MARITIME SCENE HEATMAP



VISUAL: SYNTHETIC MARITIME EDGE DETECTION



Boat, buoy, and water texture. Sobel 3×3 amplifies wave noise. Sobel 15×15 suppresses it while preserving the horizon and objects. Canny misses the buoy entirely.

CLAIMS VS EVIDENCE

What’s wrong and what’s right

OVERREACHING CLAIMS

Claim 1: "ML is unreliable and dangerous" for autonomous vehicles

$\times$ Cites only 2010 and 2016 sources. Ignores massive advances in ML safety and robustness.

Claim 2: Industry should "divert efforts" to image processing

$\times$ Scope is edge detection in synthetic/aquatic images. Cannot generalize to all autonomous driving.

Claim 3: WVF outperforms SOTA ML models

$\sim$ True on their 4-image aquatic dataset, but $n = 4$ has no statistical power. No confidence intervals reported.

Claim 4: $<1°$ angular accuracy

$\checkmark$ Empirically demonstrated on synthetic data. Valid but lacks theoretical error bounds.

STATISTICAL CONCERNS

The 4-image aquatic dataset:

- Hand-labeled by the authors
- No inter-annotator agreement
- No confidence intervals
- No *p*-values or effect sizes
- ODS/OIS differences of 2–5% could be noise

Standard practice:
BSDS500 uses 500 images with multiple annotators per image.

BIPED test set concern:

DexiNed and TEED were *trained* on BIPED. Testing them on BIPED (even test split) gives them a distribution advantage.

The WVF/LF results on BIPED are therefore *less* impressive, not more.

UDED results are more informative (out-of-distribution), but margins are only 1–3%.

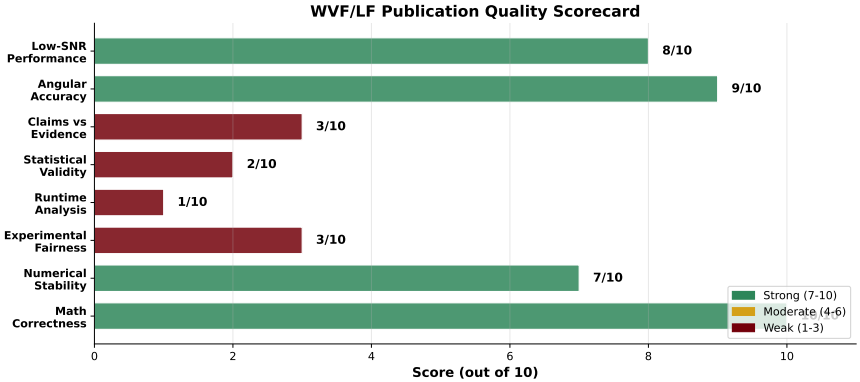
Missing baselines:

- No Gaussian-smoothed Sobel
- No LoG at matched scale

OVERALL ASSESSMENT

Fair assessment of the contributions

PUBLICATION QUALITY SCORECARD



Math is excellent (10/10). **Angular accuracy** and **low-SNR performance** are genuine strengths. **Experimental fairness**, **runtime analysis**, and **statistical validity** are the major weaknesses.

STRENGTHS & WEAKNESSES

Strengths

1. Orientation-specific gradients via rotated Taylor expansion—**genuinely novel**
2. Cubic spline angle detection—**elegant**, demonstrably superior to arctan
3. Low-SNR aquatic performance fills a **real gap**
4. WVF + LF are complementary (point vs line edges)
5. Multi-scale GMM framework is a meaningful extension

Weaknesses

1. Only compared to 3×3 Sobel/Prewitt—**unfair**
2. No runtime analysis—**critical omission**
3. $n = 4$ aquatic dataset—**no statistical power**
4. WVF is 10,000$\times$ slower—**impractical**
5. Claims about ML “unreliable and dangerous”—**unsupported**
6. Significant text overlap between the three publications

RECOMMENDATIONS

For the authors:

1. Compare against Sobel at matched spatial scale (9×9, 15×15)
2. Publish GPU-optimized implementation with timing benchmarks
3. Expand aquatic dataset to $\geq$50 images with independent annotators
4. Report confidence intervals on all metrics
5. Narrow claims to angular accuracy advantage—the genuine contribution
6. Remove unsubstantiated claims about ML safety

The WVF's real niche:

Orientation-specific gradient computation for angular accuracy in noise.

Larger Sobel kernels can match ODS/OIS, but they **cannot** provide:

- Sub-degree angular precision
- Orientation-selective noise rejection
- Wave direction estimation

Focus the narrative here.

Thank You

Code & Results:

`https://github.com/j-vaught/edge-detection-filter-critique`

All benchmarks run on Theia HPC (NVIDIA A100 GPUs)
University of South Carolina